Tewari's Space Power Generator test

Does the published loaded test show an efficiency above one? The energy ledger of UPSM pp. 159–160 — register entry H13

kronspice simulation report · 2026-11-07 · published test values (UPSM §6.7)

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Summary

P. Tewari's Space Power Generator (SPG) is a homopolar machine: an electromagnet on a rotating iron core, a static yoke, and output brushes on slip rings, with the Faraday-disk EMF V = ½ωBR² (UPSM eq. 6.16; ω the angular speed, B the field, R the radius). His book reports one loaded test and reads it as an efficiency above one. This example runs that test, number by number, through the energy ledger.

- **The arithmetic holds** apart from a typo: 3.41 V × 3438 A = 11 724 W (eq. 6.22 prints 11 727 W); the ratios 2.065 and 2.49 follow.
- **Counting all inputs** (drive motor 8620 W + excitation 650 W = 9270 W) the claimed ratio falls to 1.26; the book leaves the excitation out.
- **The whole surplus rests on one assumption:** that the EMF under 3438 A load equals the 3.41 V measured at no load. Energy balance with the book's own shaft-power figures allows at most 1.37 V (1.65 V even with a lossless drive motor).
- **Torque:** the drive motor delivered 16.7 N·m extra under load; the claimed 3.41 V at 3438 A needs 41.5 N·m of back torque.
- **Status: untested.** The book reports no loaded-EMF measurement. The decisive measurement is the sum of voltage drops around the loaded loop, or a torque transducer on the shaft.

The claim and the standard prediction

Hypothesis (H13, SPG part). Tewari's reactionless generator delivers more electrical power than the mechanical power that drives it; the book calls the result a violation of energy conservation (p. 161).

Standard prediction. The mechanical power the rotor takes from the shaft equals E_{load}·I (Poynting's theorem; the back torque is k·I when E = k·ω, wherever the reaction lands). Under load the EMF falls through armature reaction, saturation and speed sag at the rotor: the load loop carries 3438 ampere-turns against the excitation.

The published test (UPSM pp. 159–160)

A dc drive motor (DM) runs the SPG at 2700 r/min.

condition	reported
no load	DM input W_0 = 2943 W; excitation 650 W; EMF E = 3.41 V
loaded, same speed	I = 3438 A; DM input W_L = 8620 W

From these the book computes:

eq.	quantity	value
6.20	$W_r = W_L - W_0$	5677 W
6.21	$W_E = E{\cdot}I$	11 724 W
6.22	incremental efficiency	2.066
6.23	SPG alone, after dividing by the DM's efficiency 0.83	2.49

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The ledger

1. Arithmetic

3.41 V × 3438 A = 11 724 W; eq. 6.22 prints 11 727 W, a typo. The ratios 2.065 and 2.49 follow.

2. All inputs

DM 8620 W + excitation 650 W = 9270 W in, against the claimed 11 724 W out: ratio 1.26. The excitation is left out of the book's ratio.

3. The assumption

W_E multiplies the load current by the EMF measured at *no load*. The book's own numbers give the extra shaft power under load: 0.83 × 5677 W = 4712 W, a torque increase of 16.7 N·m at 282.7 rad/s. Since the rotor's shaft power must equal $E_{load}\cdot I$, this bounds the loaded EMF:

	loaded EMF	back torque at 3438 A
standard prediction	**≤ 1.37 V (≤ 1.65 V even with a lossless DM)**	**16.7 N·m, as the drive motor delivered**
the claim	3.41 V, unchanged from no load	needs k·I = 41.5 N·m

So the standard prediction is that the EMF falls by at least 60 % at 3438 A. The implied loop resistance E_0/I is 0.992 mΩ. The book reports no loaded-EMF measurement; it says the drops “can also be” measured (p. 160).

Validation

- *main_test.go* reproduces the book's numbers and the bound (1.3706 V, and 1.6513 V with a lossless drive motor), with torque consistency to 10^{-12}.
- *main.go* holds the published values and the analysis of eqs. 6.20–6.23, all inputs and the loaded-EMF bound.

Limits of the analysis

- It uses only the published figures; the shaft power under load is inferred from the drive motor's input and the book's assumed efficiency 0.83, not measured.
- It is a ledger of one reported test, not a field model of the machine.
- Whether the shaft bearings could supply the missing torque is asked in *examples/bearing_motor*; the register records that bearing (Huber) torque could close the 24.8 N·m torque gap only in EM mode, whose back-EMF leaves the same 1.37 V, and is excluded in thermal mode at the measured η ≈ 0.10. The electron model of the same hypothesis is on the bench in *examples/space_vortex*.

Status

Register status: **untested**. The claim rests on an unmeasured loaded EMF: ≤ 1.37 V predicted, 3.41 V assumed. **Decisive measurement, not in the book:** the sum of the voltage drops around the loaded loop (brushes, rings, conductor, shunt, load) at 3438 A, or the shaft torque from a transducer rather than inferred from the drive motor's input and an assumed efficiency. A closed-loop, self-running test would settle it outright.

Reproducing the results

```
go run ./examples/space_power_generator
go test ./examples/space_power_generator
```

Sources: P. Tewari, *Universal Principles of Space and Matter*, Grain and Chaff, Toronto, 2017 (1st ed. 2009), §6.7, pp. 149–161 (hypothesis source); P. Tewari, Physics Essays 31, 108 (2018); reading notes in doc/tewari-references.md.

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